

VSS Scoreboard Overlay Official Guide

System Introduction

- Prepared for: **Varsity Media Foundation**
- Project: **VSS Live Streaming Tools for OBS**
- Created By: **Hsuan-Jun (Raymond) Lin**
 - Contact Info: hlin152@asu.edu
- Last Update: **Apr 02, 2026**

Overview

This official guide explains how to:

- Implement VSS Scoreboard Overlay into OBS
- Implement VSS Scoreboard Control Panel with local server host
- Game Status will update in real-time while broadcasting

Supported Sports Type:

1. **Football**
2. **Basketball**
3. **Lacrosse**
4. **Volleyball**
5. **Hockey**

*The Scoreboard is designed as an HTML structure and using Browser Source in OBS

Required Files

Before starting, make sure you have:

- **VSS_Scoreboard.html***
- **VSS_Scoreboard_Control_Panel.html** (Optional)
- **config.json** (Initially empty; It will require to read the data for both Overlay and Control Panel)
- **Photo** Folder

New Feature 2026/04/02: Separate Control Panel with Custom Browser Docks link with local server host

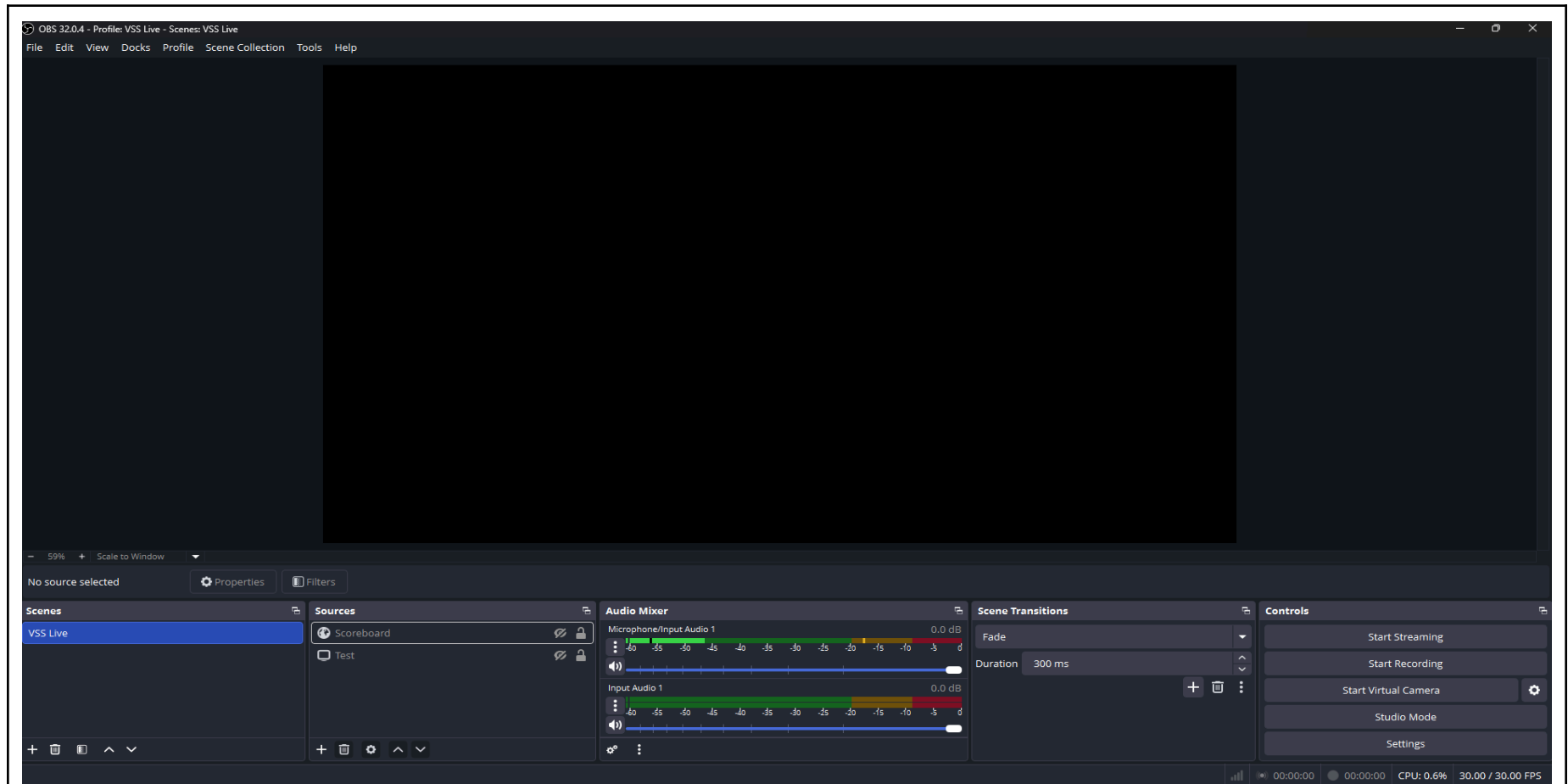
- **node_modules** Folder
- **VSS_Live_Start.bat***
- **package.json***
- **package-lock.json***
- **server.js***
- **OBS_Game.html***
- **OBS_Setup.html***

Github Download Link:

- ★ Prototype: [Scoreboard_Prototype_version3.0](#)
- ★ History Develop Scoreboard Overlay and Control Panel: [Github Update](#)
- ★ New Feature System: [VSS_OBS.zip](#)

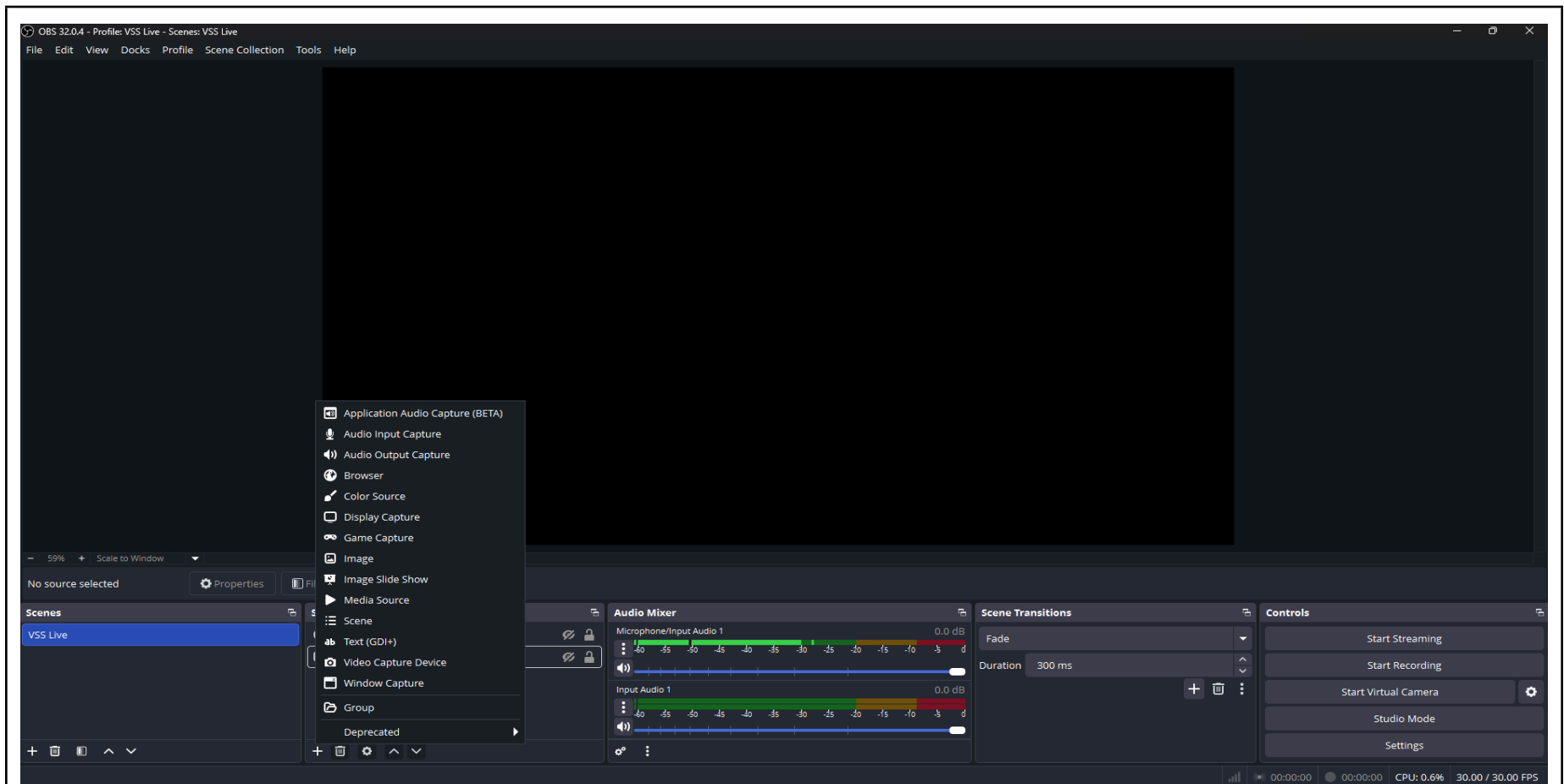
Implement VSS Scoreboard Overlay in OBS Guide

★Step1 - Open OBS

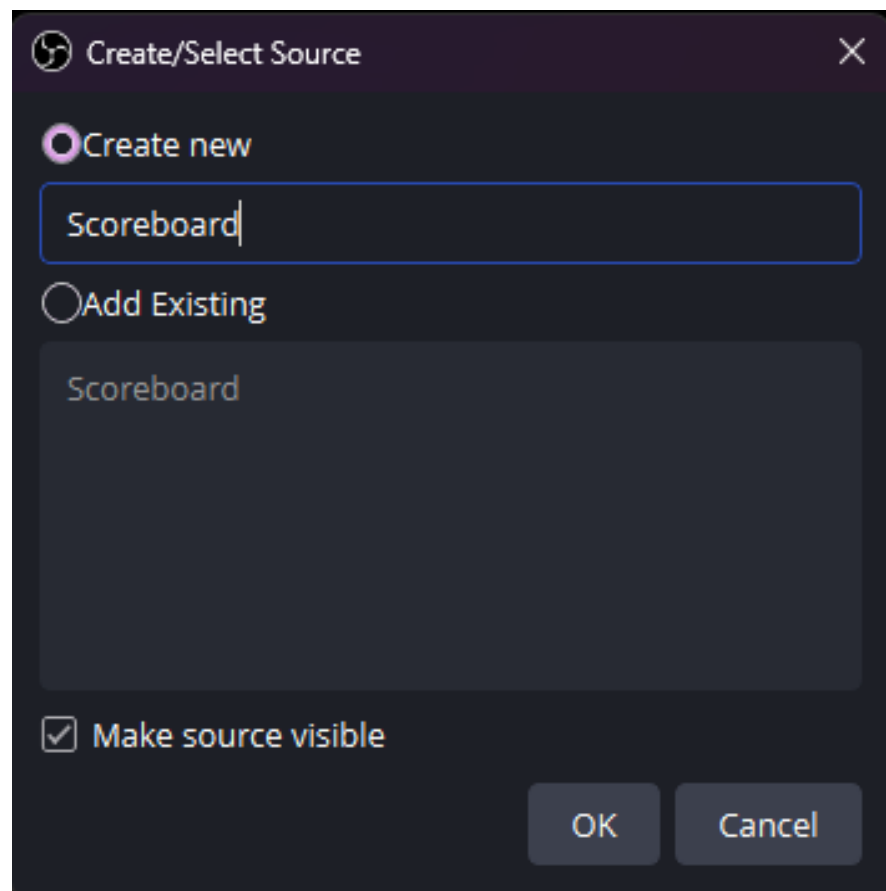


➤ *Launch OBS Studio*

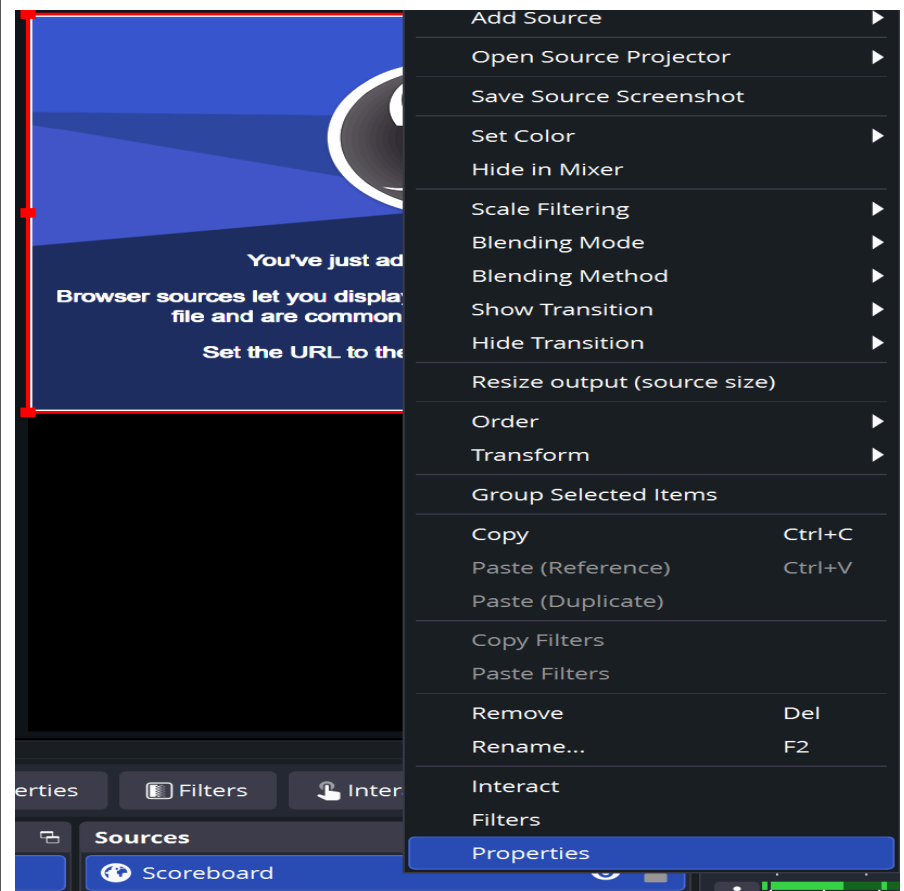
★Step2 - Setup Browser Source



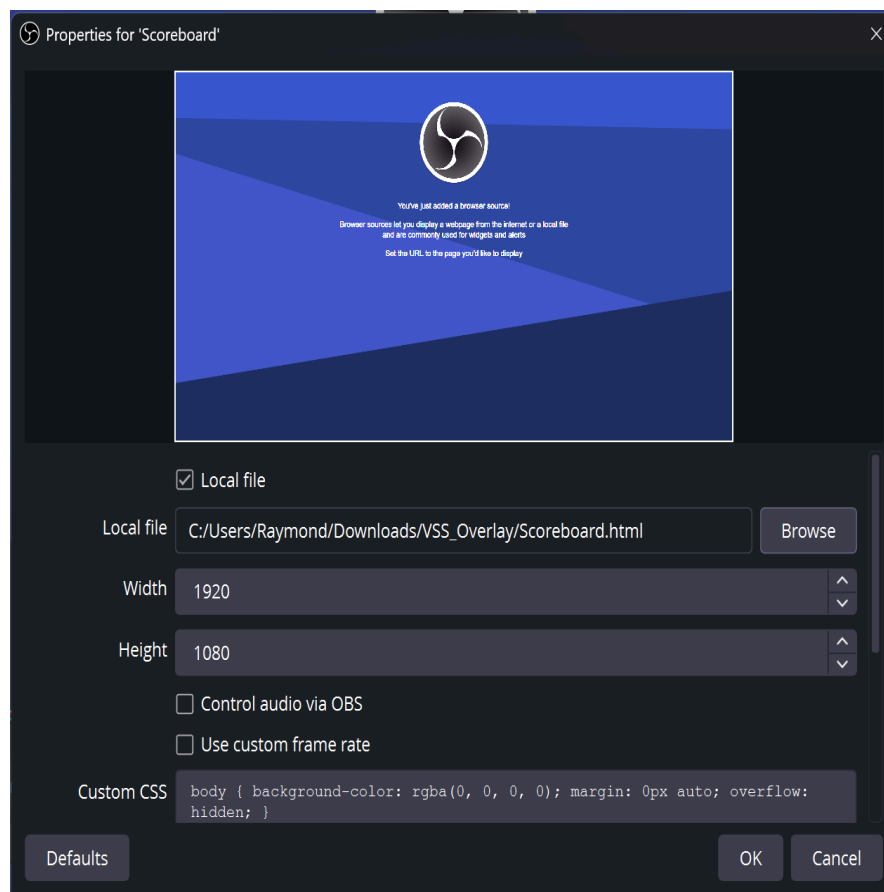
- In the **Sources** section, click the + button
- Select **Browser**



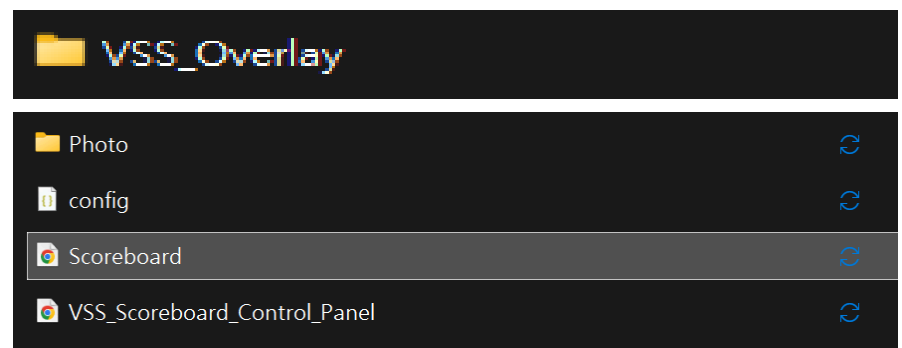
- *Rename the Source name to your preferred name (e.g., Scoreboard)*
- *Click **OK***



- *In the **Sources** section, right-click the source*
- *Select **Properties***

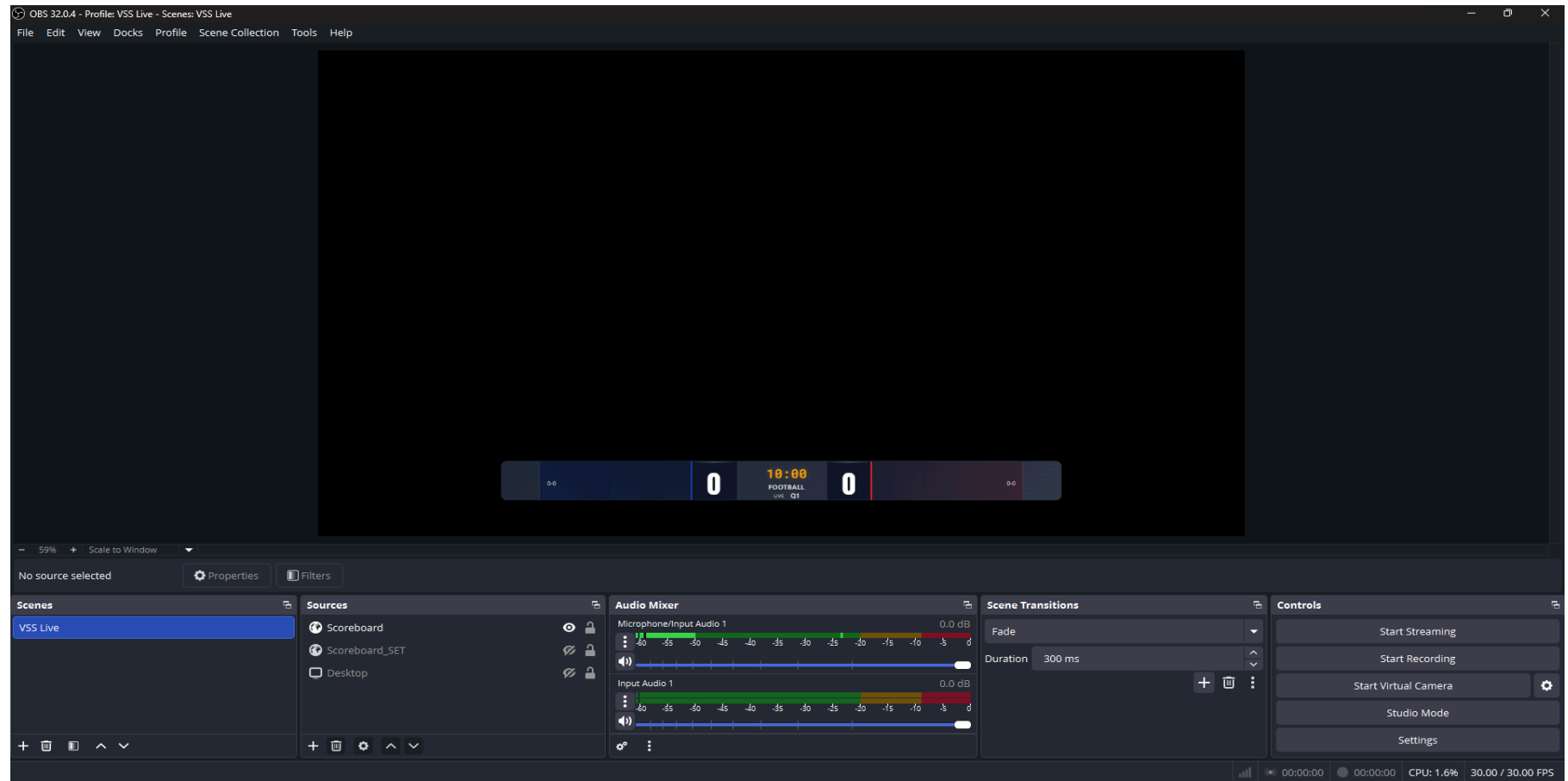


- **Select the *Local File* checkbox**
 - **Click *Browse***
- (See the next column for reference)*
- **Set the preferred resolution:**
Width: 1920



- **Browse to the *VSS_Overlay* folder**
- **Select the *Scoreboard.html* file (Update new version if needed)**

Height: 1080
(1920 × 1080 is the recommended standard)
➤ Click **OK**



➤ **Display the Scoreboard Template in OBS**

★Step3 - Setup Scoreboard Control Panel

Change Project Folder

Current folder: VSS_Over1ay

Config: config.json

Write status: OK (typing)

Clock: STOP

Version: 3.1

Auto write

Write now

Reset All

VSS Team Info

Team A Name

e.g. SCPS

Team B Name

e.g. SCOTTSDALE USD

Team A Color

Team B Color

Team A Logo

e.g. Photo/teamA.p

Browse

Team B Logo

e.g. Photo/teamB.p

Browse

Sponsor Logo (VSS)

e.g. Photo/sponsor.

Browse

League Logo

e.g. Photo/league.p

Browse

★ Place logo image files inside the Photo folder
(Photo must be in the same directory as Scoreboard.html)

VSS Game Info

TEAM RECORD

Record A

Win

-

0

+

Lose

-

0

+

Record B

Win

-

0

+

Lose

-

0

+

Reset Record A

Reset Record B

TEAM SCORE

Score A

-

0

+

Score B

-

0

+

Reset Scores

★ Record and Score changes update the scoreboard in real time

VSS Game Status

Game Clock (MM:SS)

10:00

Start / Pause

Reset Time

Game Status

LIVE

Game Period

Q1

★ The Clock runs the game time when you press [Start/Pause]
★ Type the timer and remember to press [Reset Time] to make it default

VSS Live Setting

Sports Type

Football

Live Game Date (Phoenix time)

2026/02/16

12

00

AM

★ Scoreboard Sports Type supported:
Football / Basketball / Lacrosse / Volleyball / Hockey

Control Panel version: 3.1

Change Project Folder

Current folder: VSS_Overlay

Config: config.json

Write status: OK (load)

Clock: STOP

Version: 4.4

Auto write

Write now

Reset All

VSS Team Info

TEAM	NAME	COLOR
Home	Northwest Christian Academy	
Visitor	Arizona State University	

NAME SCALE 25px

LOGO	FILE
Home	Photo/teamA.png
LOGO SCALE 10px	
Visitor	Photo/teamB.jpg
LOGO SCALE 10px	
Sponsor (VSS)	Photo/VSS.jpg
LOGO SCALE 10px	
League	Photo/league.png
LOGO SCALE 10px	

★ Place logo image files inside the Photo folder
 (Photo must be in the same directory as Scoreboard.html)

VSS Game Status

TEAM RECORD & SCORE

TEAM	WINS / LOSSES
Northwest Christian Academy	5 2
Arizona State University	2 5

RECORD SCALE 25px

★ Record and Score changes update the scoreboard in real time

TEAM	SCORE
Northwest Christian Academy	92
+6 -6 +3 -3 +1 -1	
Arizona State University	114
+6 -6 +3 -3 +1 -1	

CLOCK CONTROLS

CLOCK	TIME	HIDE/SHOW
Game Clock	10:00	
Shot Clock	40	

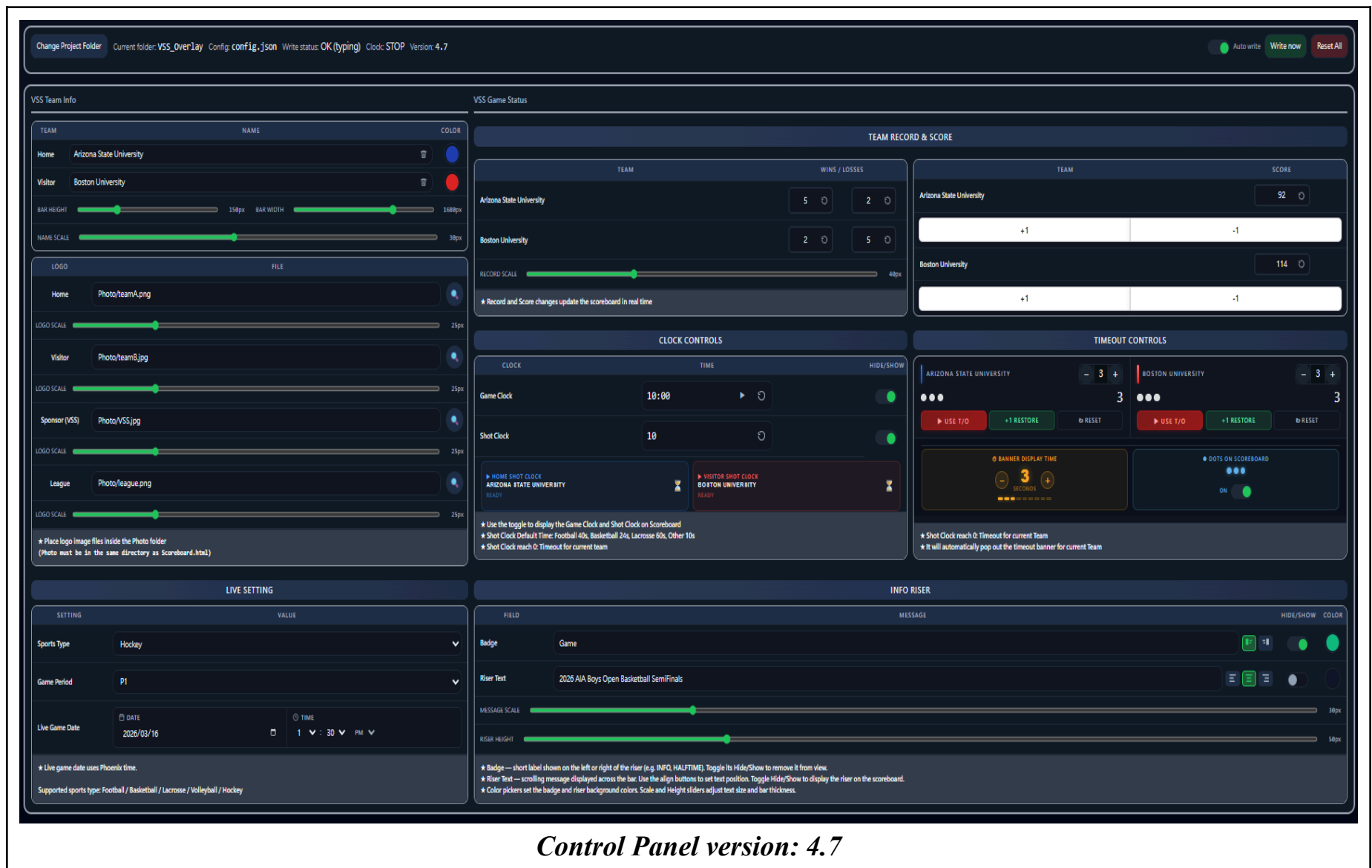
★ Once entering the clock time, press Start button to define the Reset time.
 Use the toggle to display the game clock and shot clock on Scoreboard

LIVE SETTING

SETTING	VALUE
Sports Type	Football
Game Period	Q1
Live Game Date	DATE 2026/03/16 TIME 1 : 30 PM

★ Live game date uses Phoenix time.
 Supported sports type: Football / Basketball / Lacrosse / Volleyball / Hockey

Control Panel version: 4.4



Control Panel version: 4.7

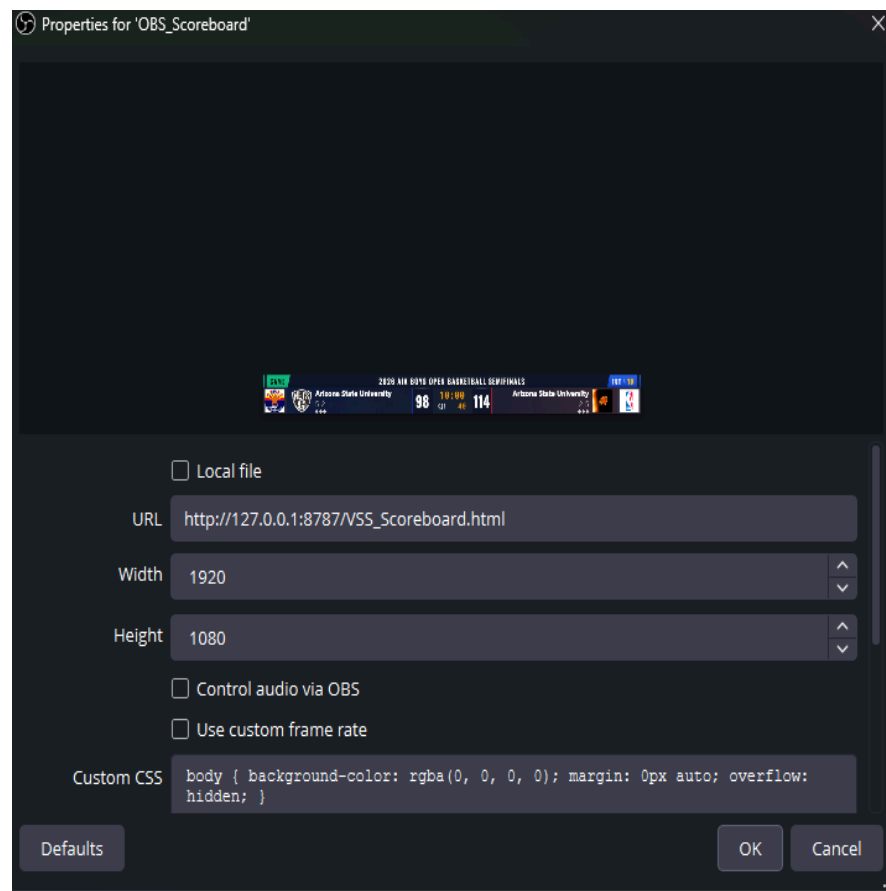
1. Open the *VSS_Overlay* folder
2. Open the File on any browser (e.g., google chrome, safari...etc.):
VSS_Scoreboard_Control_Panel.html
Make sure to use the Latest version of Control Panel to use newest feature and UI experience
3. Click *Change Project Folder*
4. Select the *VSS_Overlay* folder as the default directory
(There is no need to navigate inside the folder).
5. Click *Allow* when prompted
6. Fill in the required information under:
 - a. *VSS Team Info*
 - b. *VSS Game Status*
 - c. *VSS Game Info (remove and combine into VSS Game Status)*
 - d. *VSS Live Setting (remove and combine into VSS Game Status)*

You may update this information during during the livestream or broadcast as needed
7. Custom images (e.g., team logos) must be placed inside the *Photo* folder within the *VSS_Overlay / VSS_OBS* directory.

Real-Time Update Behavior

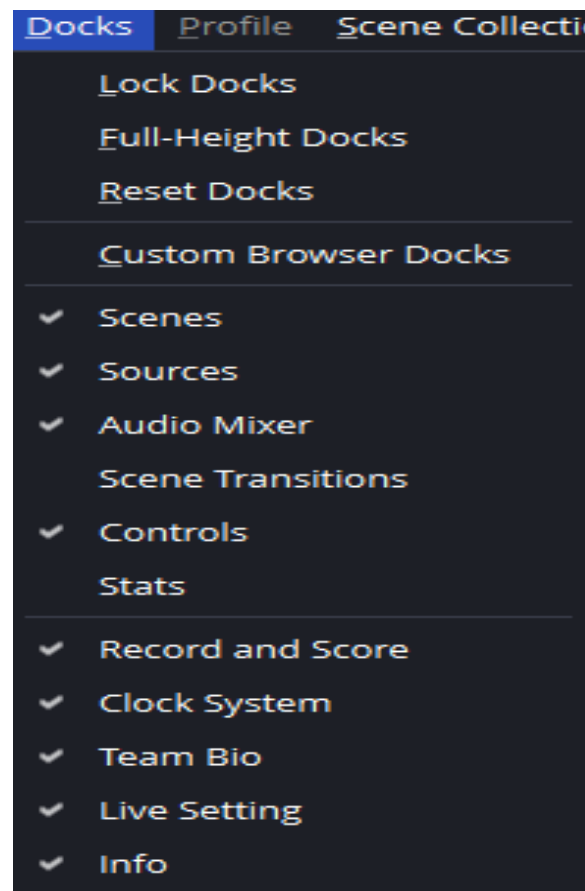
- The control panel will automatically perform real-time updates synchronized with the OBS Scoreboard.
- If you toggle **Auto Write OFF**, you will need to manually click “**Write Now**” each time to apply updates.
- By default, **Auto Write is ON**, and no changes are required.

2026/04/02 New Implementation with local server host on OBS



- **Unselect *Local file***
- **File URL:**

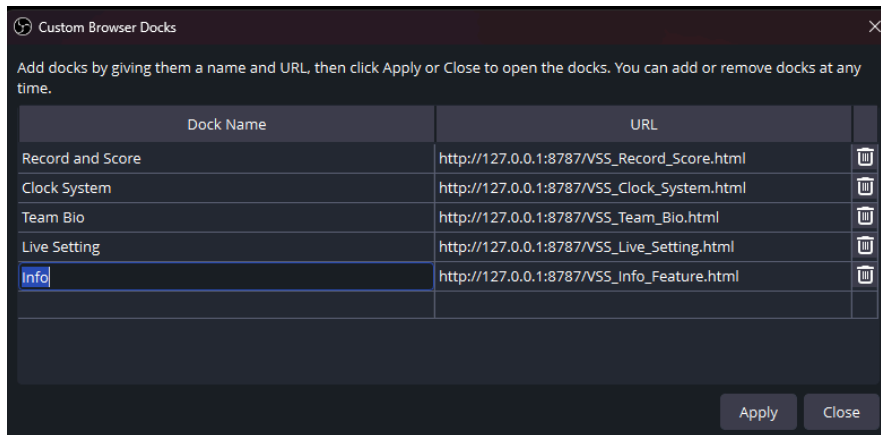
http://127.0.0.1:8787/VSS_Scoreboard.html



- **In the *Docks* section on top**
- **Select *Custom Browser Docks***

➤ **Click *OK***

***Check above Guide if needed**



- Enter any **Dock Name** you like
- make sure these url is included for all component

http://127.0.0.1:8787/OBS_Setup.html

http://127.0.0.1:8787/OBS_Game.html

- **Apply**
- **Customize your own display way on OBS**

New Feature File:

- **node_modules Folder***
- **VSS_Live_Start.bat***
- **server.js***
- **package.json***
- **package-lock.json***
- **OBS_Setup.html***
- **OBS_Game.html***

***: Mandatory file to operate in OBS**

Requirement:

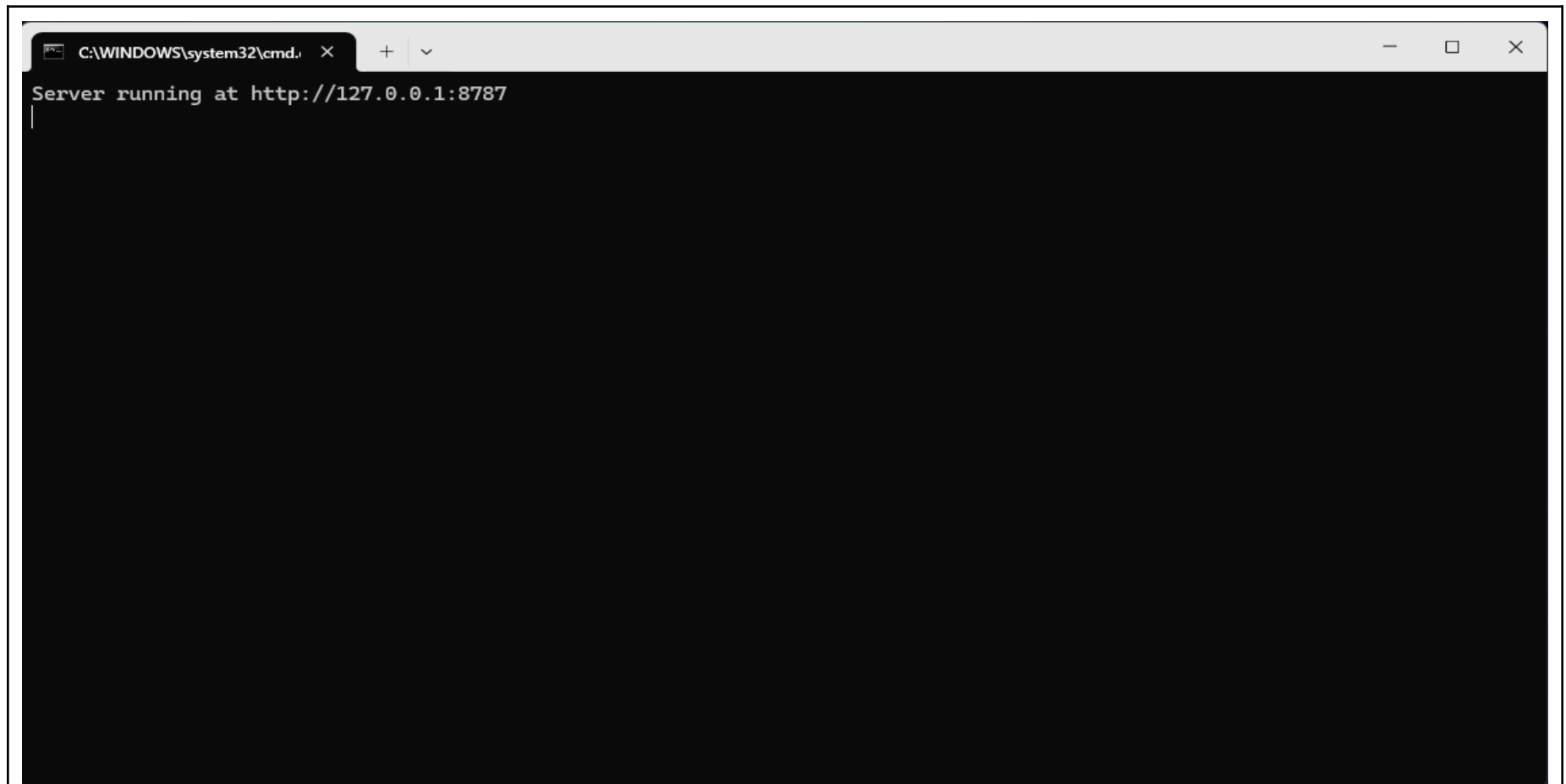
- Download Node.js on Windows: [Node.js](https://nodejs.org/en/download/)

*If Broadcaster already have it on pc then skip it

- Download file for local server host: [VSS_OBS.zip](#)

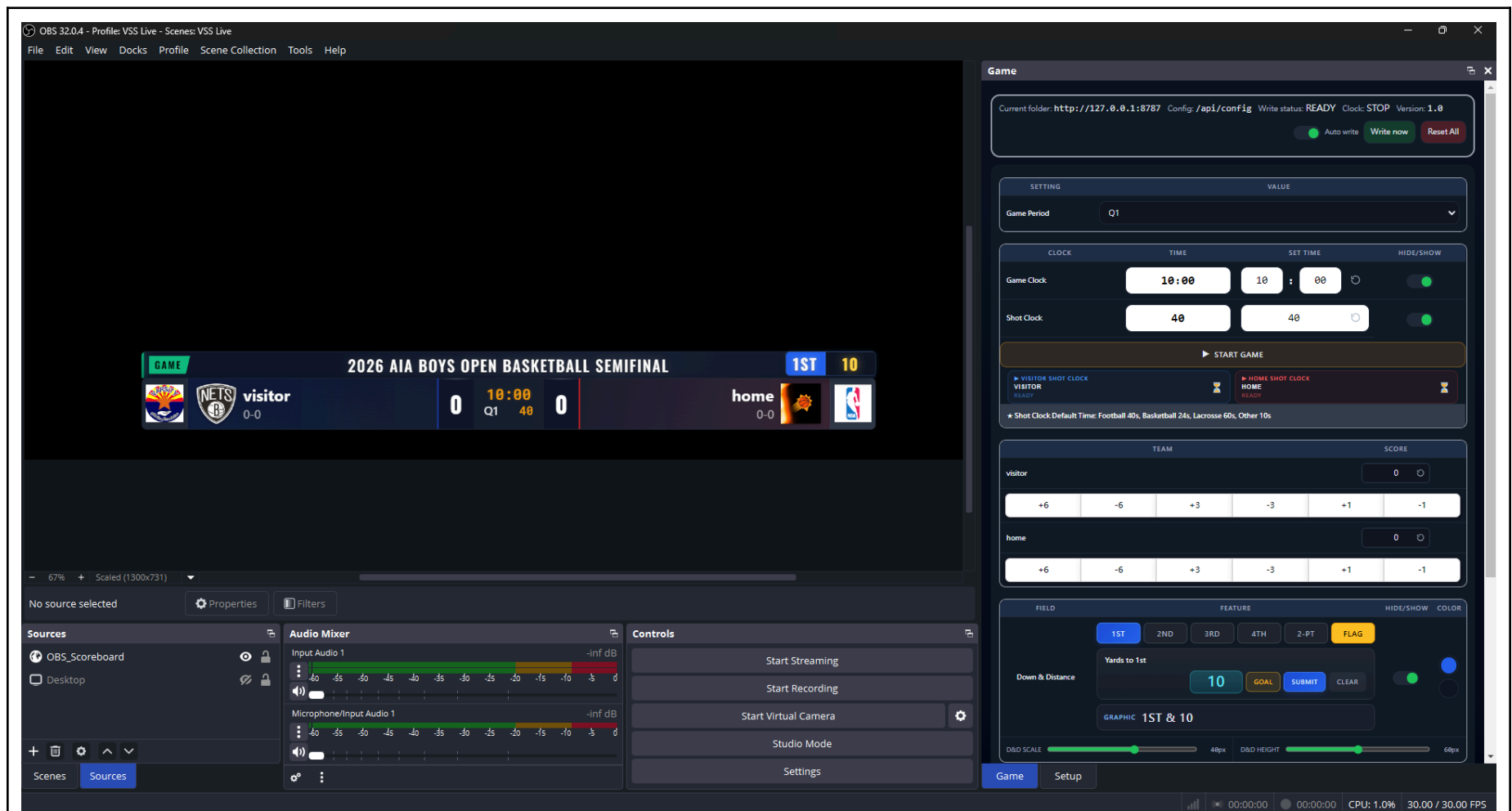
Server Host on local device:

- Once OBS is fully setup, open **VSS_Live_Start.bat**

A screenshot of a Windows command prompt window. The title bar at the top shows the path 'C:\WINDOWS\system32\cmd...' and standard window controls (minimize, maximize, close). The command prompt itself has a black background with white text. The first line of text reads 'Server running at http://127.0.0.1:8787'. A vertical cursor is positioned at the end of this line, ready for the next command.

```
C:\WINDOWS\system32\cmd...  
Server running at http://127.0.0.1:8787  
|
```

***DO NOT** Close this local server while broadcasting, it will stop sync the data in real-time



Recommend Setup Example:

OBS_Setup Docks & **OBS_Game Docks** can overlap in same Docks section. Using switch field to change display.

Development Adjustment Suggestion Board

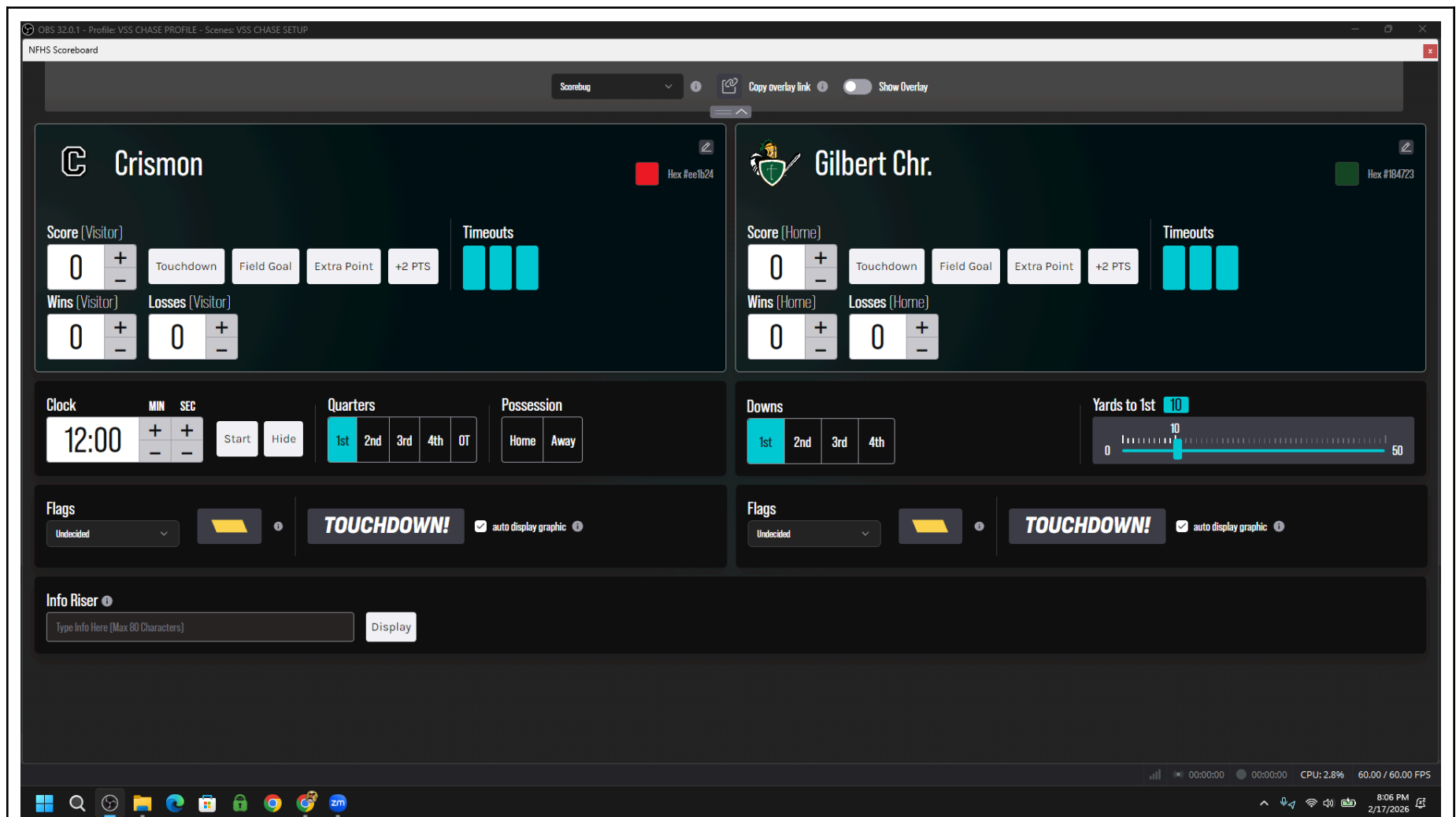
Enter any suggestion that needs to change or adjust (Feature, UI, BUG, Development...etc.):

Completed Feature: check Development / Completed Update Log below

Chase Notes:

I really love how responsive the control panel is with the graphic is so great job there!

Here is a screengrab of the NFHS scoreboard control panel for football to give an example of the functionality needed:



Here is a recording to the full functionality of the NFHS Football scoreboard. It also has penalty indicators, timeout

animations, and possession indicators. Hopefully all this is of help to give you guys some more ideas on what you can do for these scoreboards!

Chase Notes 3/6/2026:

- Nice job with this update! A lot of things that are really good here. Keep up the good work!
- The logo scaling option doesn't allow you to go as big as I think it should be. It would be nice if you could uncap the limit of the sizing
- Play clocks/shot clocks are still needed in the sports that have them
- A way to show and hide the clocks would also be nice to implement so we can hide it if the time is off and don't want viewers to see
- Having the scoreboard say "football" on it is not necessary. If someone is on the stream they will already know the sport. If that could be removed and have period/quarter made a little larger in its place that would be great
- I think the game clock could also be made a bit bigger to fill in the empty space of the box that it is in as well

Chase Notes 3/16/2026

- Some scaling issues need to be addressed. When scaling one logo and not the other, it moves both of them (see screenshot). This could be fixed by making sure everything remains centered vertically as you scale.
- The shot clock itself works quite well but its just too big on the scoreboard. Could you move the shot clock to the red x and the quarter/period to the blue O. This way its not blocking a large part of the screen with that box. (Screenshot Below)
- This would help address the centering problem as seen below as well. All the graphics and scores should be centered vertically on the blue line as shown on the second screenshot below.
- There is also currently a lot of dead space on the scoreboard that could be either cut out or adjustable with a slider. The red area of the screenshot could be eliminated with a width slider but then if one team has a longer name it could extend back out to accommodate

- An info riser box would be nice to have at the top of the scoreboard. Something that could be shown and hidden to include some text that the operator can change. I was thinking about the size and location of the green box below. I also included an example of something we would put there. Having the box scale to the text would be nice as well.

Take a look at popular scorebugs attached below for a reference on spacing and info that should be on the scoreboards



Vince Question 3/16/2026— can the scoreboard be managed within the OBS screen itself, or is it controlled from the html web browser? Would be easier to manage if the controls were in OBS. How do I add the control panel (above) to OBS so I can control the scoreboard there?

Chase Notes 3/30/2026:

- The shot clock placement is much better, and I really like the UI for it however it needs to be able to be paused if one team keeps possession after it goes out of bounds. I like it resetting still when you activate the shot clock for the other team, though.
- The centering vertically for the clocks still isn't quite aligned when on the shortest scoreboard height. We talked about this during the call that there needs to be more padding on the bottom
- Adding in a down and distance section for the football scoreboard is a must. This will be a variable text element that changes between 1st, 2nd, 3rd, and 4th downs, followed by the yardage left to gain a first down (up to 99 yards). The format for the text goes as follows: 1st & 10, 3rd & 8, 4th & 2, 2nd and 15. The UI should allow the user to set the down then type in the distance. The distance should not auto update on the graphic. There should be a "submit" that allows us to have the correct info before finalizing it on the graphic. Some other things that should also be implemented into this down and distance box is text saying 2-point att. There should also be an option to put letters in the distance section for when there are goalline situations. These would appear like so: 2nd & GOAL, 4th & GOAL.

For more details on down and distance, please take a look at the attached article. It has all the basics of what you need to know in order to make this work.

<http://dearsportsfan.com/2011/07/05/whats-a-down-in-football-ive-been-pretending-to-know-but-i-dont/>

- UI cleanup request: one big thing that would help would be to have separate pages on the control panel so that it isn't all cluttered onto one page. One page should have all of the general info about team names, their colors, pictures, and graphic sizing. A different page should have only the things that we would need to control during the game. These would be the score, clocks, quarters, info riser, and timeouts. Something that could also declutter the control panel would be to make the scaling bars for each element a drop down tab so it can hide away until needed.

- Implementation of colors being able to change for text and backgrounds on each part of the graphic and other locations would be greatly appreciated

Development / Completed Update Log

Raymond Notes:

UI could be updated to have Scoring presets for each sport example, +6, +1, and +3 for football, or +3, +2, and +1 increments for basketball so that we don't need to count up one at a time.

Completed:

Integrate the Scoring Presets in Control Panel version 3.3

The text for the team names and records are really small and tough to read if those could b/e resized. Also for team names if text could scale to accommodate longer team names like "Northwest Christian Academy" so that it will fit well that would be great.

Completed:

Integrate the Team Name and Team Record Font Size UI in Control Panel version 3.4

Integrate the Team Name and Team Record Font Size function in Scoreboard version 3.2

I don't see a reason to have the LIVE text next to the Q1 text

Completed:

remove Game Status in Control Panel version 3.3

remove Game Status in Scoreboard version 3.1

I would really like the ability to resize or reframe the images in the scorebug if possible. If the image I have is too small, I am going to have to go to another program and resize the image then reupload it into the scorebug.

Completed:

Integrate the Scale to resize the image in Control Panel version 4.0

Integrate the Scale to resize the image optimize in Scoreboard version 4.0

Optimize the newest Modern UI to easily control and have better visualization for use

The logo scaling option doesn't allow you to go as big as I think it should be. It would be nice if you could uncap the limit of the sizing

Having the scoreboard say "football" on it is not necessary. If someone is on the stream they will already know the sport. If that could be removed and have period/quarter made a little larger in its place that would be great

I think the game clock could also be made a bit bigger to fill in the empty space of the box that it is in as well

Completed:

hidden sports type, optimize period and game clock bigger on scoreboard template in Scoreboard version 4.1

logo scaling uncap to 60px in Scoreboard Control Panel version 4.1

(If scaling over normal size, the scoreboard can't match the properly format)

*2026/03/11 Raymond Note

Also needed are play clocks that can run alongside game clocks. Basketball, football, and lacrosse, to name a few. These shot clocks are crucial game info. These clocks should also have their own toggle to show and hide.

Play clocks/shot clocks are still needed in the sports that have them

Completed:

add the shot clock in Scoreboard version 4.3

add the shot clock feature and optimize the UI in Scoreboard Control Panel version 4.3

*2026/03/12 Raymond Note

A way to show and hide the clocks would also be nice to implement so we can hide it if the time is off and don't want viewers to see

Completed:

allow hide the clocks time and adjust the period to the middle in Scoreboard Control Panel version 4.4

optimize the better visualization for use in Scoreboard Control Panel version 4.4

add OT option for each sports game period in Scoreboard Control Panel version 4.4

*2026/03/12 Raymond Note

Chase Notes 3/16/2026

The shot clock itself works quite well but its just too big on the scoreboard. Could you move the shot clock to the red x and the quarter/period to the blue O. This way its not blocking a large part of the screen with that box. (Screenshot Below)

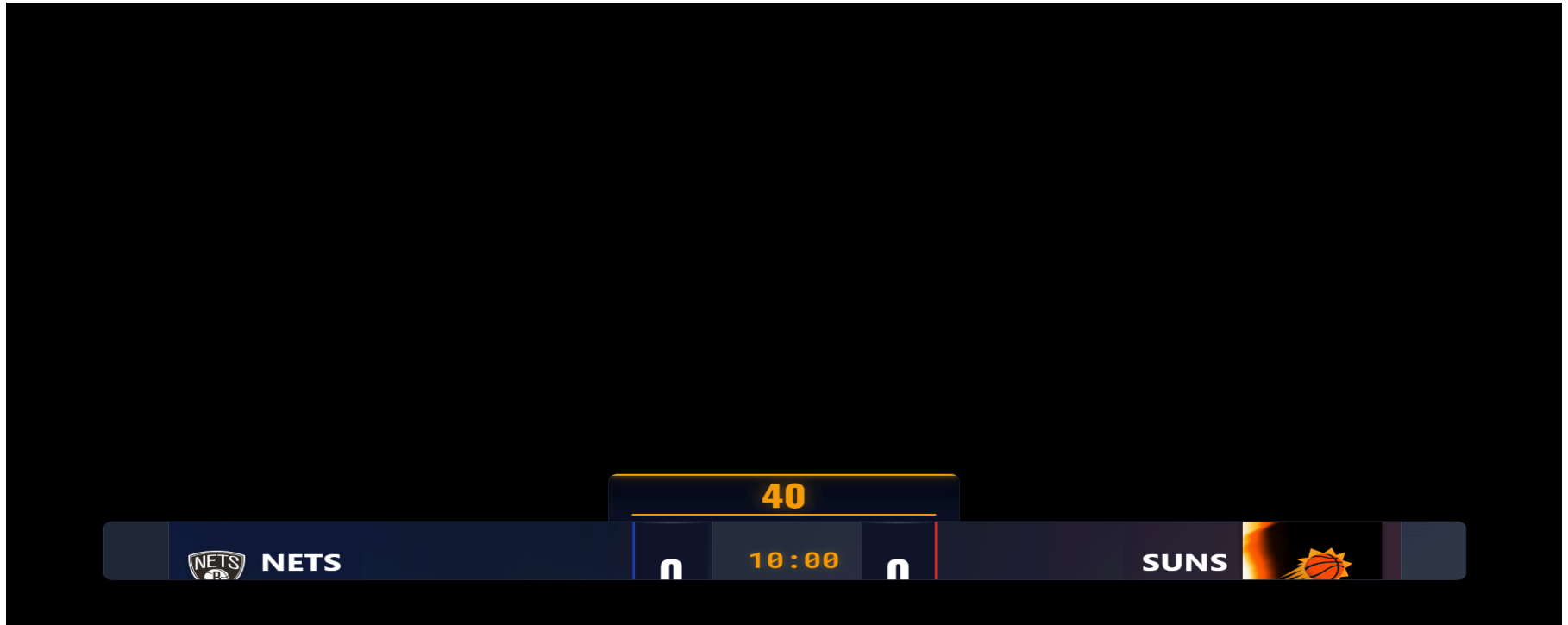


Completed:
adjustment period to left and shot clock to right in Scoreboard version 4.4
*2026/03/16 Raymond Note

Chase Notes 3/16/2026

This would help address the centering problem as seen below as well. All the graphics and scores should be centered vertically on the blue line as shown on the second screenshot below.

There is also currently a lot of dead space on the scoreboard that could be either cut out or adjustable with a slider. The red area of the screenshot could be eliminated with a width slider but then if one team has a longer name it could extend back out to accommodate



NETS

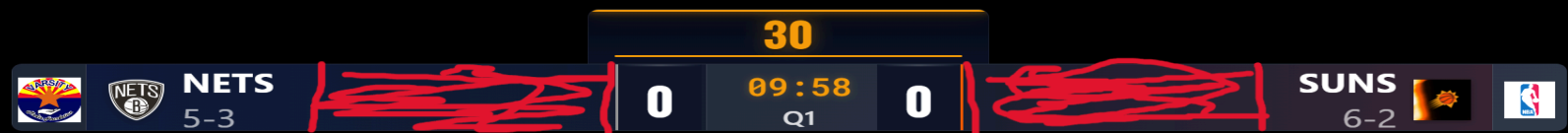
0

10:00

0

SUNS





Completed:

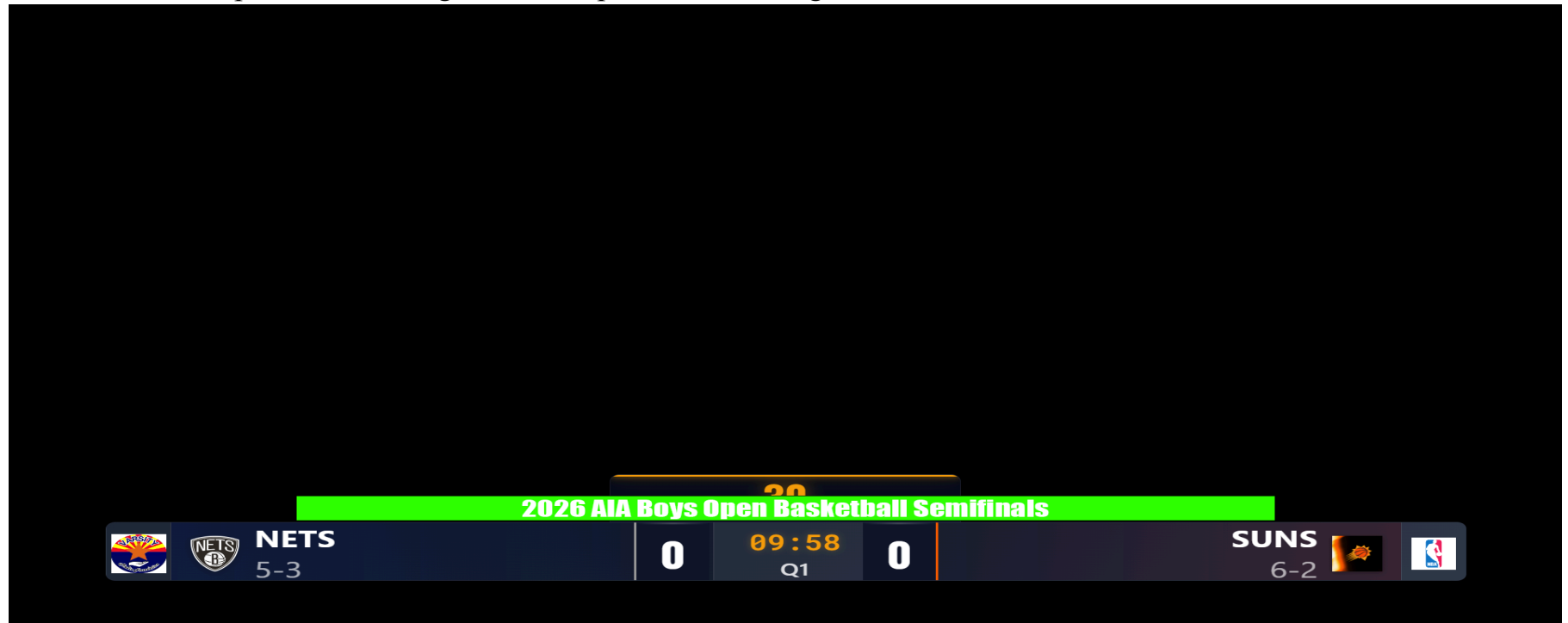
Scaling issue fix in Scoreboard version 4.5

Scaling issue fix and add the Scoreboard width and height customized in Scoreboard Control Panel version 4.5

*2026/03/16 Raymond Note

Chase Notes 3/16/2026

An info riser box would be nice to have at the top of the scoreboard. Something that could be shown and hidden to include some text that the operator can change. I was thinking about the size and location of the green box below. I also included an example of something we would put there. Having the box scale to the text would be nice as well.



GAME

2026 AIA BOYS OPEN BASKETBALL SEMIFINALS



Arizona State University
5-2

92

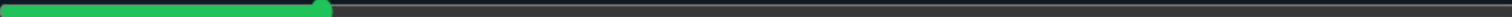
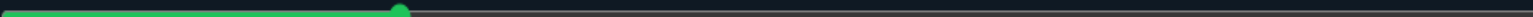
10:00
Q1 40

114

Arizona State University
2-5



INFO RISER

FIELD	MESSAGE	HIDE/SHOW	COLOR
Badge	Game	   	
Riser Text	2026 AIA Boys Open Basketball SemiFinals	   	
MESSAGE SCALE			30px
RISER HEIGHT			50px

★ Badge — short label shown on the left or right of the riser (e.g. INFO, HALFTIME). Toggle its Hide/Show to remove it from view.

★ Riser Text — scrolling message displayed across the bar. Use the align buttons to set text position. Toggle Hide/Show to display the riser on the scoreboard.

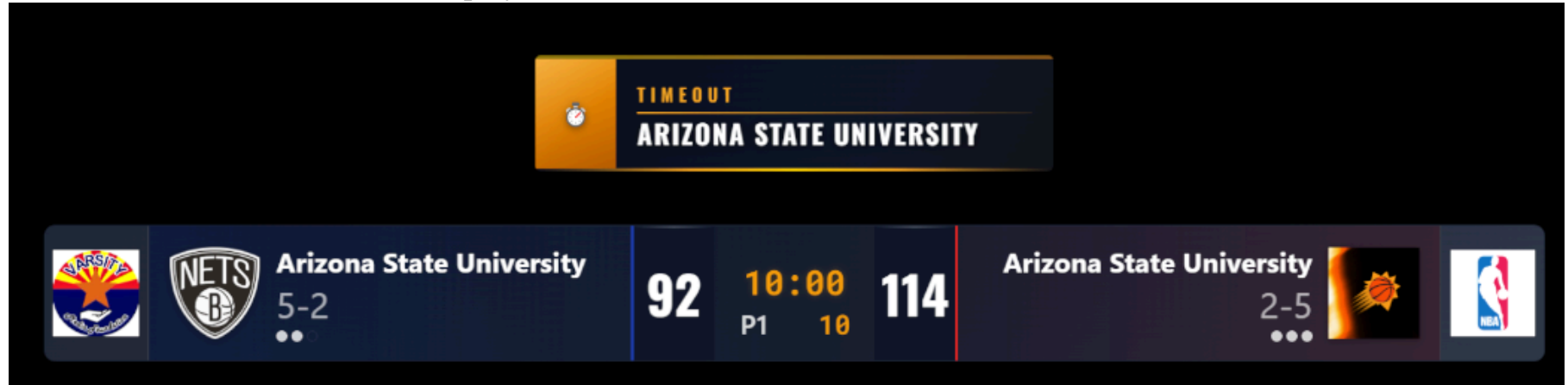
★ Color pickers set the badge and riser background colors. Scale and Height sliders adjust text size and bar thickness.

Completed:
Add Info Riser Feature in Scoreboard version 4.6
Add Info Riser Feature in Scoreboard Control Panel version 4.6

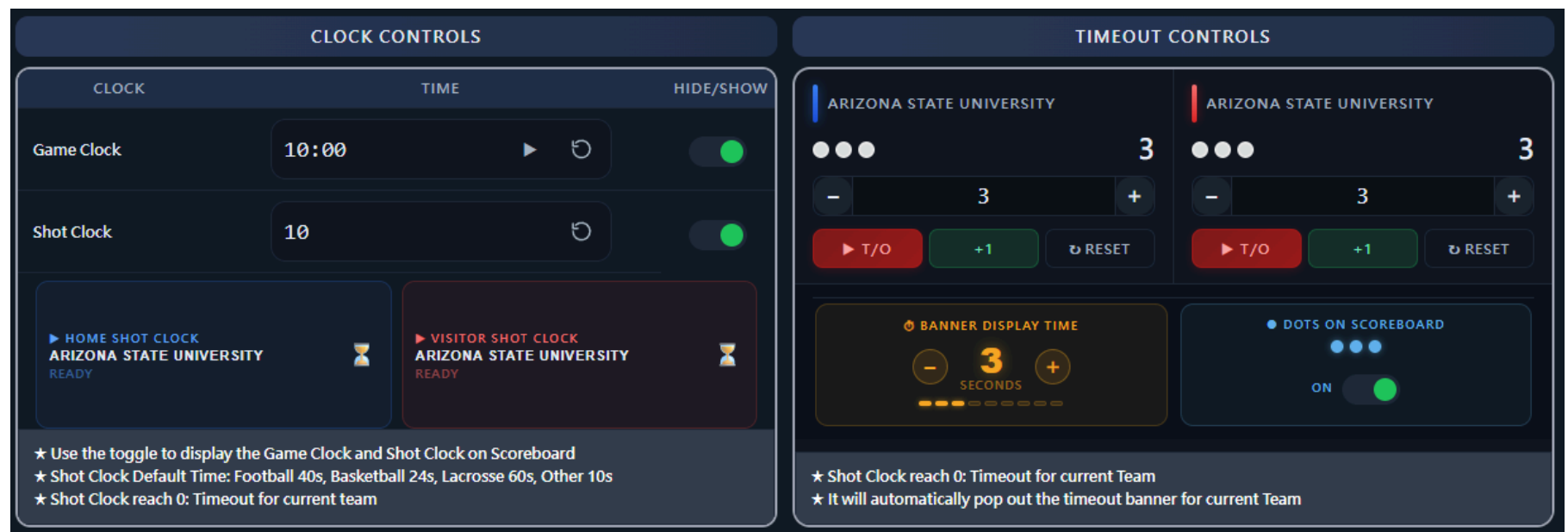
(Check the detail on above screenshot, user can customized the color, text align, text message, scaling...etc)

*2026/03/16 Raymond Notes

Timeout feature on Scoreboard Display



Timeout feature on Scoreboard Control Panel



Completed:

Add Timeout Feature and adjust shot clock logic in Scoreboard version 4.7

Add Timeout Feature and adjust shot clock logic in Scoreboard Control Panel version 4.7

*2026/03/17 Raymond Note

Vince Notes 3/16/2026

can the scoreboard be managed within the OBS screen itself, or is it controlled from the html web browser? Would be easier to manage if the controls were in OBS. How do I add the control panel (above) to OBS so I can control the scoreboard there?

Completed:

New Designed Docks is available through VSS_OBS.zip

*2026/04/02 Raymond Note

Chase Notes

A show/hide toggle for the graphic with an animation to make it appear and disappear on OBS

Scoring animations that play for each type of score. A TOUCHDOWN! Animation for when a touchdown is scored. A +3 or +2 animation whenever baskets are scored for basketball.

TODO:

not available in this stage

Chase Notes 3/30/2026:

- The shot clock placement is much better, and I really like the UI for it however it needs to be able to be paused if one team keeps possession after it goes out of bounds. I like it resetting still when you activate the shot clock for the other team, though.
- The centering vertically for the clocks still isn't quite aligned when on the shortest scoreboard height. We talked about this during the call that there needs to be more padding on the bottom

Completed:

Adjust several logic in Scoreboard version 4.8

Adjust several logic in Scoreboard Control Panel version 4.8

*2026/03/31 Raymond Note

Chase Notes 3/30/2026:

- Adding in a down and distance section for the football scoreboard is a must. This will be a variable text element that changes between 1st, 2nd, 3rd, and 4th downs, followed by the yardage left to gain a first down (up to 99 yards). The format for the text goes as follows: 1st & 10, 3rd & 8, 4th & 2, 2nd and 15. The UI should allow the

user to set the down then type in the distance. The distance should not auto update on the graphic. There should be a “submit” that allows us to have the correct info before finalizing it on the graphic. Some other things that should also be implemented into this down and distance box is text saying 2-point att. There should also be an option to put letters in the distance section for when there are goalline situations. These would appear like so: 2nd & GOAL, 4th & GOAL.

Completed:

Integrated Football down and distance section in Scoreboard version 4.9

Integrated Football down and distance section in Scoreboard Control Panel version 4.9

* Download the Latest Version of both file from Github and update the previous patch

* Require to reload the new Scoreboard into OBS if needed

*2026/03/31 Raymond Note

Chase Notes 3/30/2026:

- UI cleanup request: one big thing that would help would be to have separate pages on the control panel so that it isn't all cluttered onto one page. One page should have all of the general info about team names, their colors, pictures, and graphic sizing. A different page should have only the things that we would need to control during the game. These would be the score, clocks, quarters, info riser, and timeouts. Something that could also declutter the control panel would be to make the scaling bars for each element a drop down tab so it can hide away until needed.

Completed:

Introduce the New HTML structure for Custom Browser Docks on OBS

- Check below

*2026/04/01 Raymond Note

Chase Notes 3/30/2026:

- Implementation of colors being able to change for text and backgrounds on each part of the graphic and other

locations would be greatly appreciated

TODO:

not available in this stage

Chase Notes 4/1/2026: *URGENT*

- The home team should be on the right of the graphic and the visitor should be on the left.
- *There are a few issues we're running into with updating the graphic. See the attached video for example.

 2026-04-01 21-31-58.mp4

What I am referring to:

- The Down and Distance resets almost all of the graphic and the only way to recover it is to use the scaling bars
- Running the clocks also seems to be resetting the graphic
- The game clock and the down and distance graphic aren't able to be activated at the same time
- Any time the down and distance is submitted, it resets both clocks to their default times
- The score gets instantly reset while the clock is running

(All this makes me think that the game clock is just resetting everything every time it runs

- The color dropper when selecting a color does not work currently
- *I like the browser dock option but it needs to be simplified. Here is a mockup of how I would like the browser docks laid out:

SETUP

The SETUP interface is a dark-themed control panel for configuring a sports scoreboard. It includes sections for:

- General Settings:** Sports Type (Football), Live Game Date (min/dd/yyyy, 12:00 AM).
- Team Configuration:** Home (Suns) and Visitor (Nets) teams with color selection and logo upload (Photo/teamB.jpg, Photo/teamA.png).
- Visual Customization:** Sliders for Bar Height, Bar Width, Name Scale, Logo Scale, and Record Scale.
- Additional Features:** Sponsor (VSS) and League logos, a message field (e.g., "Halftime — 2026 AIA Boys Basketball"), and riser text.

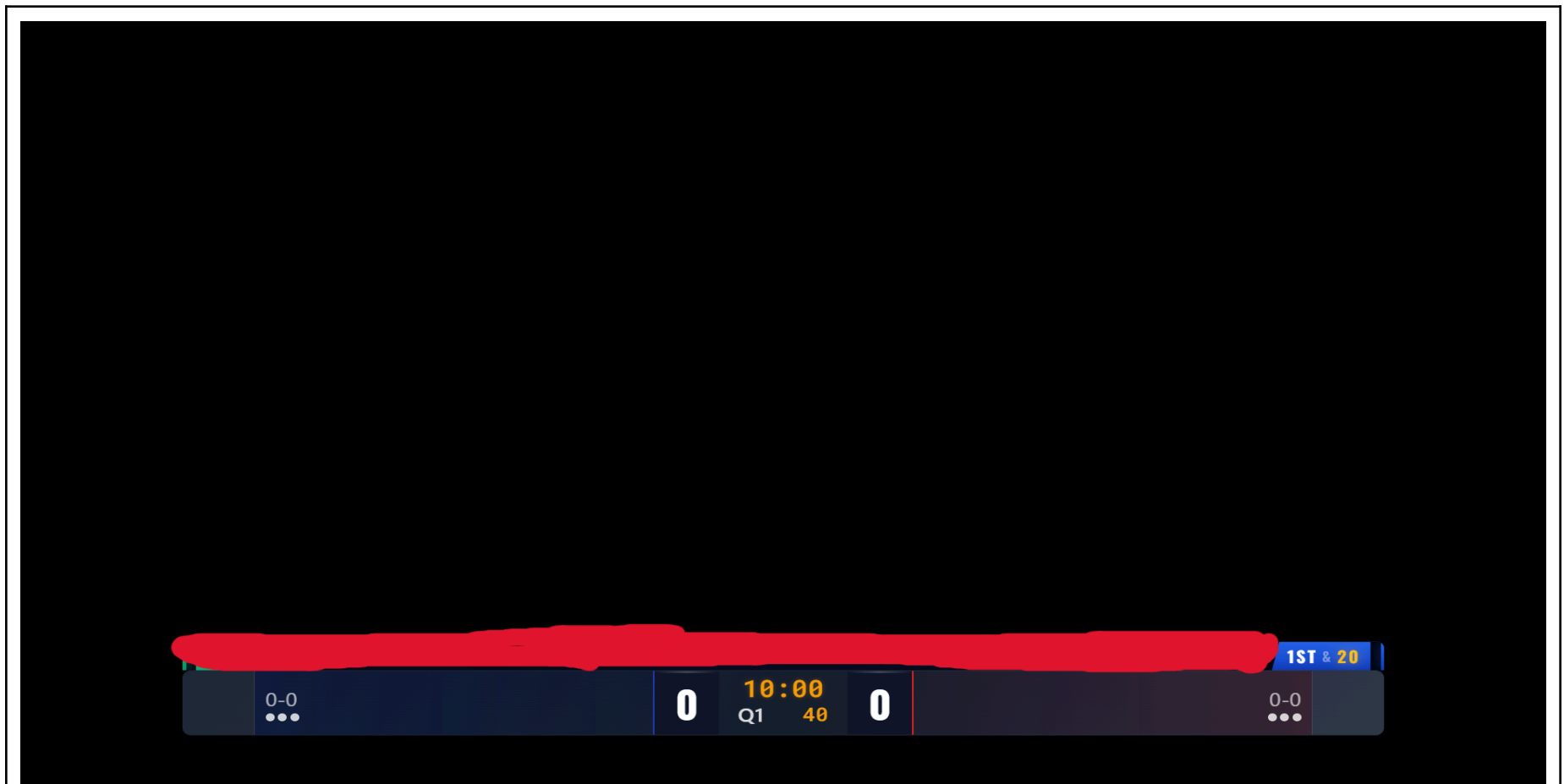
GAME

The GAME interface is a dark-themed live scoreboard display. It includes sections for:

- Game Information:** Game Period (Q1), Game Clock (09:46), and Shot Clock (40).
- Scoreboard:** Home (Suns) and Visitor (Nets) scores, with a "START GAME" button.
- Gameplay Controls:** "Down & Distance" section showing "Yards to 1st" (20) and a "SUBMIT" button.
- Scoreboard Details:** Individual player scores (e.g., 3, 3, 3) and a "T/O" (Time Out) button.
- Additional Features:** "BANNER DISPLAY TIME" (3 SECONDS) and "DOTS ON SCOREBOARD" (ON/OFF toggle).

This allows us to consolidate everything into two concise browser docks.

- *We need the option to put the down and distance up by itself. The red area should be toggleable if we want it there for text or not.



Completed:

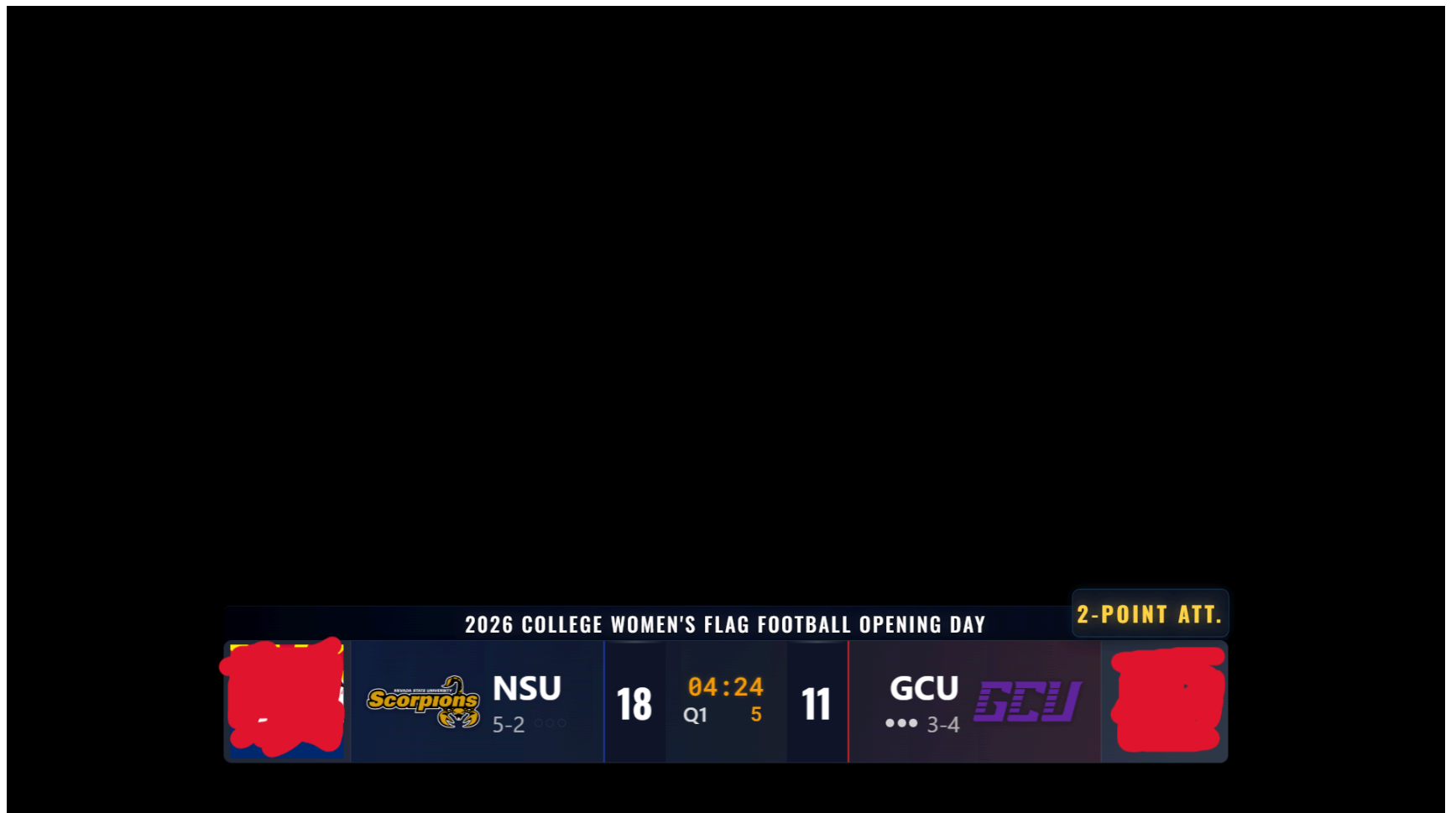
Fix UI display and relative logic issue

* Check **New Implementation with local server host on OBS** above

*2026/04/02 Raymond Note

Chase Notes 4/2/2026:

- First off, excellent job with the updated UI. It is concise and easy to follow.
- The down and distance graphic doesn't allow me to change it while the clock is running.
- When the shot clock runs out of time, it brings up the timeout graphic. Nothing should happen when that clock hits zero except it showing a zero on the clock
- Could you add a tab on the down and distance graphic that says "FLAG" with black text on a yellow background? This would use the whole box exactly how you have the 2-Point Att. set up.
- For scoring for football, the game clock should stop when you score a touchdown. Can you make it to when you select the +6 option for both teams, it stops the game clock only?
- Could you allow the red area to be able to toggle on and off? It's a lot of blank space if we don't have a league sponsor for a specific game.



Completed:

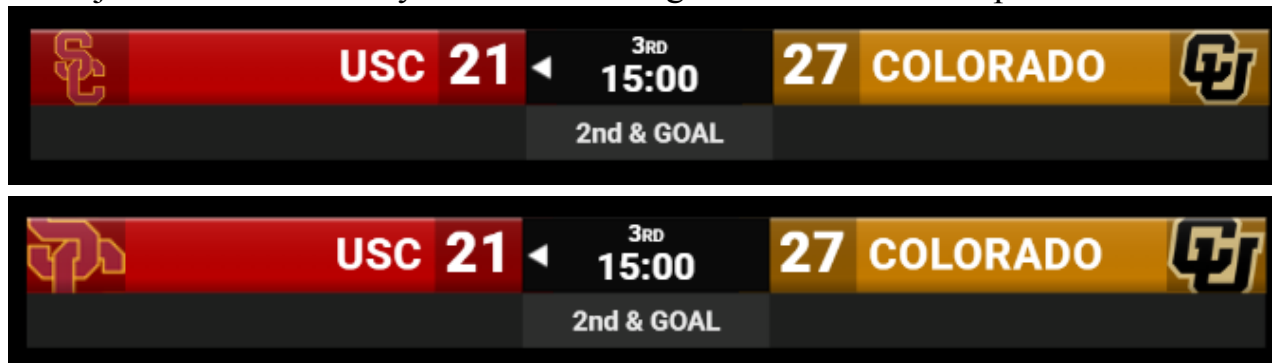
Fix UI display and relative logic issue

* Check **New Implementation with local server host on OBS** above

*2026/04/02 Raymond Note

Chase Notes 4/3/2026:

- The shot clocks should not automatically reset. They should stay on zero until either: teams are switched or the reset button is hit
- For the down and distance graphic, can you make it one color and have it say “&” between the down and distance
- Can you uncap the scaling factor for the logos that will allow us to scale the logos as shown in each of the screenshots below. These two scaling options adds a lot of flexibility with logos we can use on the scoreboard. An adjustable X-axis and y axis for these logos would also be helpful if we can zoom in more.



Completed:

1. Shot Clock will stay 0 until user press reset button or switch team shot clock
2. Down & Distance will now stay in single color with & between.
3. Uncap from 100px to 500px

* Check **New Implementation with local server host on OBS** above

*2026/04/06 Raymond Note